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SMMC 2025 B3

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Reflection

Let  $n$  be a positive integer and let  $\omega = e^{2\pi i/n}$ . Suppose  $z_1, z_2, \dots, z_n$  are distinct complex numbers. Define  $A$  to be the  $n \times n$  matrix whose  $(j, k)$ -th entry is  $(z_j - \omega^k)^n$ , for  $1 \leq j, k \leq n$ .

Given that  $\det A = 0$ , prove that  $z_1 z_2 \cdots z_n = 1$ .

# Thoughts I

## 1. Setting:

1.1  $n$  is a positive integer.

1.2  $\omega = e^{2\pi i/n}$ .

1.3  $z_1, \dots, z_n$  are distinct complex numbers.

1.4  $A = (a_{jk})$  where  $a_{jk} = (z_j - \omega^k)^n$ .

1.5  $\det A = 0$ .

## 2. Aim: $z_1 z_2 \cdots z_n = 1$ . Now we begin to brainstorm what we can do.

## 3. (1.2) tells us that $\omega$ is an $n$ -root of unity. So I should recall some useful results for me to use later.

3.1  $1, \omega, \omega^2, \dots, \omega^{n-1}$  are roots of the polynomial  $x^n - 1$ .

3.2  $1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$ .

## 4. (1.4) gives some observations:

4.1  $a_{jk}$  looks like a polynomial of degree  $n$ , e.g.,  $p(x) = (x - \omega^k)^n$ .

4.2 Applying binomial expansion gives

$$(z_j - \omega^k)^n = \sum_{i=0}^n \binom{n}{i} z_j^{n-i} (-1)^i \omega^{ki}.$$

# Thoughts II

- 5. (1.5) is equivalent to a lot of results, I write down some tools I can use.
  - 5.1  $\text{rank } A < n$  (personally do not think it is useful here)
  - 5.2 The columns of  $A$  are linearly dependent / The system  $Ax = 0$  has nontrivial solution  $x$ .
  - 5.3 The rows of  $A$  are linearly dependent.
- 6. (5.2) means there exist  $\alpha_1, \dots, \alpha_n \in \mathbb{C}$  not all are zero such that  $\sum_k \alpha_k \cdot (\text{kth column of } A) = 0$ .
- 7. Breakdown the equation in (6): If we look at the  $j$ th row, we see that  $\sum_k \alpha_k (z_j - \omega^k)^n = 0$  for  $j = 1, \dots, n$ .
- 8. (★) The LHS of (7) looks like a polynomial of degree  $n$ , i.e.,  $p(x) = \sum_k \alpha_k (x - \omega^k)^n$ . Also, under this setting of  $p(x)$  together with (7),  $z_1, \dots, z_n$  are the roots of the polynomial  $p(x)$ .

## Thoughts III

9. (8) maybe has some issue: we assume that  $p(x)$  is a polynomial of degree  $n$ .
10. I am not confident yet to (8) since expanding the expression we see that the coefficient of  $x^n$  is  $\alpha_1 + \cdots + \alpha_n$ . We can settle it if  $\alpha_1 + \cdots + \alpha_n \neq 0$ .
11. I do not need to show (10). From (1.3), we have  $n$  distinct roots for  $p(x)$ . So it guarantees  $p(x)$  is of degree  $n$ . It clears the doubt in (9).
12. From (11), we see that  $\alpha_1 + \cdots + \alpha_n \neq 0$  (which is the leading coefficient).

## Thoughts IV

13. (★) (12) and (2) lead me to recall Vieta's formula: Given a polynomial  $q(x) \in \mathbb{C}[x]$  of degree  $n$  with roots  $r_1, \dots, r_n$ , we have  $r_1 r_2 \cdots r_n = (-1)^n \frac{\text{Constant}}{\text{Leading coefficient}}$ .
14. We can apply the formula in (13) based on our context:
- 14.1  $r_i = z_i$  for all  $i$ . So  $r_1 r_2 \cdots r_n = z_1 z_2 \cdots z_n$ .
- 14.2 The leading coefficient of  $p(x)$  is  $\sum_k \alpha_k = \alpha_1 + \cdots + \alpha_n \neq 0$  (argued in (12)).
- 14.3 The constant term is  $p(0)$ . By (8) we obtain  $p(0) = \sum_k \alpha_k (-1)^n \omega^{kn}$ .
- 14.4 Combining (14.1) until (14.3) we see that

$$z_1 z_2 \cdots z_n = (-1)^n \frac{\sum_k \alpha_k (-1)^n \omega^{kn}}{\sum_k \alpha_k} = \frac{\sum_k \alpha_k \omega^{kn}}{\sum_k \alpha_k}.$$

## Thoughts V

15. So it remains to show  $\frac{\sum_k \alpha_k \omega^{kn}}{\sum_k \alpha_k} = 1$ .
16. I stuck for a while and just notice that (3) tells me  $\omega^n = 1$ .
17. So (14.3) can be improved:  $\omega^{kn} = (\omega^n)^k = 1$  which implies  $\frac{\sum_k \alpha_k \omega^{kn}}{\sum_k \alpha_k} = 1$ .
18. After reviewing arguments, I notice a new issue: (9)-(11) only works if  $p(x)$  is a NONZERO polynomial. We need to ensure that the polynomial constructed in (8) itself is not a zero polynomial.
19. I cannot proceed anything from the construction. Maybe I should add one condition so I can continue: Suppose for the sake of contradiction that  $p(x) = 0$  for all  $x \in \mathbb{C}$ .

## Thoughts VI

20. By (4.2) (replace  $z_j$  by  $x$ ), we see that

$$\begin{aligned} 0 &= p(x) \\ &= \sum_k \alpha_k \sum_{i=0}^n \binom{n}{i} x^{n-i} (-1)^i \omega^{ki} \\ &= \sum_k \sum_{i=0}^n \alpha_k \binom{n}{i} x^{n-i} (-1)^i \omega^{ki} \\ &= \sum_{i=0}^n \sum_k \alpha_k \binom{n}{i} x^{n-i} (-1)^i \omega^{ki} \\ &= \sum_{i=0}^n \binom{n}{i} x^{n-i} (-1)^i \sum_k \alpha_k \omega^{ki}. \end{aligned}$$

21. Comparing the coefficient of  $x^{n-i}$ , we see that  $\binom{n}{i} (-1)^i \sum_k \alpha_k \omega^{ki} = 0$  for all  $i = 0, 1, \dots, n$ .

## Thoughts VII

22.  $\binom{n}{i}(-1)^i$  is nonzero.
23. By (21) and (22) we get  $\sum_k \alpha_k \omega^{ki} = 0$  for all  $i = 0, 1, \dots, n$ .
24. (★) (23) can be described using matrix: Let  $B = (b_{ik})_{0 \leq i \leq n, 1 \leq k \leq n}$  be a  $(n+1) \times n$  matrix given by  $b_{ik} = \omega^{ki}$ . Then  $B\alpha = 0$ .
25. Notice that
- ▶ when  $i = 0$ , we have  $\sum_k \alpha_k \omega^{ki} = \sum_k \alpha_k$ ;
  - ▶ when  $i = n$ , we have  $\sum_k \alpha_k \omega^{kn} = \sum_k \alpha_k$  (in case forgot, see argument in (17)).
26. (25) can be used to improve (24): Now  $B = (b_{ik})_{1 \leq i \leq n, 1 \leq k \leq n}$  is an  $n \times n$  matrix given by  $b_{ik} = \omega^{ki}$  and so  $B\alpha = 0$ .

## Thoughts VIII

27. The matrix  $B$  is

$$\begin{pmatrix} \omega^{1 \cdot 1} & \omega^{2 \cdot 1} & \dots & \omega^{n \cdot 1} \\ \omega^{1 \cdot 2} & \omega^{2 \cdot 2} & \dots & \omega^{n \cdot 2} \\ \omega^{1 \cdot 3} & \omega^{2 \cdot 3} & \dots & \omega^{n \cdot 3} \\ \vdots & \vdots & \ddots & \vdots \\ \omega^{1 \cdot n} & \omega^{2 \cdot n} & \dots & \omega^{n \cdot n} \end{pmatrix} = \begin{pmatrix} \omega^{1 \cdot 1} & \omega^{2 \cdot 1} & \dots & 1 \\ \omega^{1 \cdot 2} & \omega^{2 \cdot 2} & \dots & 1 \\ \omega^{1 \cdot 3} & \omega^{2 \cdot 3} & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ \omega^{1 \cdot n} & \omega^{2 \cdot n} & \dots & 1 \end{pmatrix}$$

28. (★) It looks like a Vandermonde matrix

$$V := \begin{pmatrix} 1 & (\omega^1)^1 & (\omega^1)^2 & \dots & (\omega^1)^{n-1} \\ 1 & (\omega^2)^1 & (\omega^2)^2 & \dots & (\omega^2)^{n-1} \\ 1 & (\omega^3)^1 & (\omega^3)^2 & \dots & (\omega^3)^{n-1} \\ \vdots & \vdots & \ddots & \vdots & \\ 1 & (\omega^n)^1 & (\omega^n)^2 & \dots & (\omega^n)^{n-1} \end{pmatrix}$$

We can obtain this matrix by switching columns of  $B$ .

## Thoughts IX

- 29. The matrix  $V$  is certainly invertible because  $\omega^1, \omega^2, \dots, \omega^n$  are distinct. (Study Vandermonde matrix if you do not know this fact.)
- 30. (27)–(29) imply that  $B$  is invertible, but this contradicts to  $B\alpha = 0$  in (26).
- 31. Hence  $p(x)$  is a nonzero polynomial, and we can go back to (9).
- 32. We can start to write solution according to the plan:
  - 32.1 (6) and (7)
  - 32.2 Construct polynomial (8); argue it is nonzero (19)–(31).
  - 32.3 (9)–(11)
  - 32.4 Vieta's formula argument (13)–(17).

## Solution I

Since  $\det A = 0$ , there exists a nonzero column vector  $\alpha = (\alpha_1, \dots, \alpha_n)^T$  such that  $A\alpha = 0$ .

By looking at the  $j$ th coordinate of  $A\alpha = 0$ . We see that

$$\sum_k \alpha_k (z_j - \omega^k)^n = 0 \quad \text{for } j = 1, \dots, n. \quad (\text{Eq. 1})$$

Let  $p(x) \in \mathbb{C}[x]$  be a polynomial defined by

$$p(x) = \alpha_k (x - \omega^k)^n.$$

**Claim.**  $p(x)$  is a nonzero polynomial.

**Proof.** We prove by contradiction.

## Solution II

Suppose that  $p(x) = 0$ . Then

$$\begin{aligned} 0 &= p(x) \\ &= \sum_k \alpha_k \sum_{i=0}^n \binom{n}{i} x^{n-i} (-1)^i \omega^{ki} \\ &= \sum_{i=0}^n \sum_k \alpha_k \binom{n}{i} x^{n-i} (-1)^i \omega^{ki} \\ &= \sum_{i=0}^n \binom{n}{i} x^{n-i} (-1)^i \sum_k \alpha_k \omega^{ki}. \end{aligned}$$

Since  $p(x)$  is a zero polynomial, the coefficients of  $x^i$ 's are zero. In particular, we have  $\binom{n}{i} (-1)^i \sum_k \alpha_k \omega^{ki} = 0$  for all  $i = 1, \dots, n$ .

## Solution III

Since  $\binom{n}{i}(-1)^i \neq 0$ , we get

$$\sum_k \alpha_k \omega^{ki} = 0 \quad \text{for } i = 1, \dots, n. \quad (\text{Eq. 2})$$

Let  $B = (b_{ik})$  be an  $n \times n$  matrix given by  $b_{ik} = \omega^{ki}$  for all  $1 \leq i, k \leq n$ .

By (Eq. 2), we obtain  $B\alpha = 0$ . Hence  $B$  is singular.

But  $\det B$  is

$$\begin{vmatrix} \omega^{1 \cdot 1} & \omega^{2 \cdot 1} & \dots & 1 \\ \omega^{1 \cdot 2} & \omega^{2 \cdot 2} & \dots & 1 \\ \omega^{1 \cdot 3} & \omega^{2 \cdot 3} & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ \omega^{1 \cdot n} & \omega^{2 \cdot n} & \dots & 1 \end{vmatrix} = - \begin{vmatrix} 1 & (\omega^1)^1 & (\omega^1)^2 & \dots & (\omega^1)^{n-1} \\ 1 & (\omega^2)^1 & (\omega^2)^2 & \dots & (\omega^2)^{n-1} \\ 1 & (\omega^3)^1 & (\omega^3)^2 & \dots & (\omega^3)^{n-1} \\ \vdots & \vdots & \ddots & \vdots & \\ 1 & (\omega^n)^1 & (\omega^n)^2 & \dots & (\omega^n)^{n-1} \end{vmatrix}$$

which is nonzero since  $\omega^1, \dots, \omega^n$  are distinct (and the last matrix is a Vandermonde matrix).

## Solution IV

We have arrived at a contradiction. So the claim is proved  $\square$

Clearly  $p(x)$  has degree at most  $n$ .

By (Eq. 1),  $p(z_j) = 0$  for  $j = 1, \dots, n$ .

Since  $z_1, \dots, z_n$  are distinct roots of the nonzero polynomial  $p(x)$ ,  $p(x)$  has degree  $n$ .

Write  $p(x) = a_n x^n + \dots a_1 x + a_0$ .

By Vieta's formula, we get

$$z_1 z_2 \cdots z_n = (-1)^n \frac{a_0}{a_n}.$$

## Solution V

Note that

$$a_n = \sum_{k=1}^n \alpha_k \neq 0,$$

$$\begin{aligned} a_0 = p(0) &= \sum_k \alpha_k (-1)^n \omega^{kn} \\ &= (-1)^n \sum_k \alpha_k (\omega^n)^k = (-1)^n \sum_{k=1}^n \alpha_k = (-1)^n a_n. \end{aligned}$$

Thus

$$z_1 z_2 \cdots z_n = (-1)^n \frac{(-1)^n a_n}{a_n} = (-1)^{2n} = 1. \quad \square$$

## Remarks

1. The draft was my thought process while invigilating SMMC 2025. I managed to work out the steps (1)–(17) on the spot, but missed the flaw (18) until I write the actual solution.
2. Whenever you spot issues like that, you can either go back to plan, or just skip it for a bit and assume it works for now (when you are running out of time).
3. The problem is interesting because it combines several elementary techniques. However, recognizing which techniques to use requires experience (pay attention to ★). I don't think I would solve it if I don't read so many algebra books.
4. I hope this gives you a feel on how to trick yourself to use key techniques, e.g., product of roots  $\Rightarrow$  Vieta; symbols with powers  $\Rightarrow$  polynomials; many equations  $\Rightarrow$  matrices.